



Class Test 4

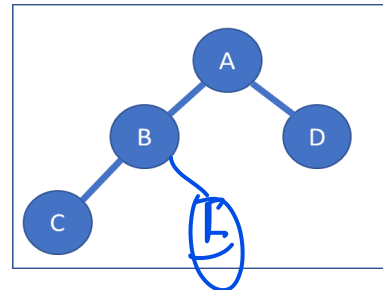
Marks : 20

Time: 20 min

- 1 Consider the given **binary tree**. Each tree node is based on the given structure template.

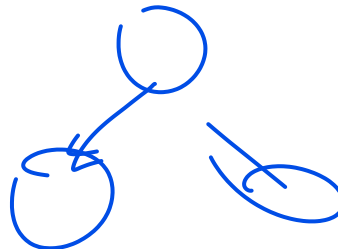
4+4

```
struct TreeNode{  
    char value;  
    struct TreeNode* parent;  
    struct TreeNode* left;  
    struct TreeNode * right;  
}
```



Write the necessary statements in order to do the following:

- Insert a node 'E' as the right child of 'B.'
- Insert a node 'F' as the left child of 'C'



2. Perform the following operations on an empty **binary search tree (BST)**.

6+6

- Insert in the BST in the following sequence **10, 15, 5, 20, 30, 18, 14, 11, 12** (draw the tree once all the elements are inserted)
- Delete the node with the value **11** (draw the tree *after each transplant operation* and also draw the *final tree*.)



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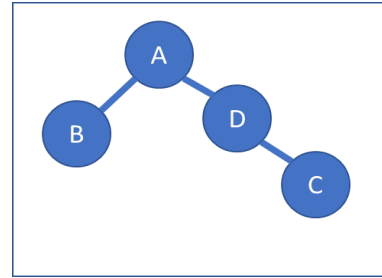
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4+4

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Write the necessary statements in order to do the following:

- Insert a node 'E' as the right child of 'B.'
- Insert a node 'F' as the left child of 'C'

2. Perform the following operations on an empty binary search tree (BST).

6+6

- Insert in the BST in the following sequence **100, 15 , 20, 30, 18, 14, 11** (draw the tree once all the elements are inserted)
- Delete the node with the value **15** (draw the tree *after each transplant operation* and also *draw the final tree.*)

